

Executive Summary & Capstone Project Report

1. **Project Objective:** The primary objective of this end-to-end case study is to analyze global financial sales data, build a predictive model for forecasting future sales, and design an interactive business intelligence dashboard. The analysis delivers actionable insights into product line performance, regional market health, and promotional effectiveness.

2. Data Cleaning & Engineering Pipeline:

- * **Whitespace Correction:** Trimmed all leading and trailing whitespaces from the structural column headers to avoid indexing issues during system cross-referencing.
- * **Missing Value Imputation:** Identified 53 missing values (NaN) within the categorical Discount Band column. These missing records were systematically imputed with a default string value of 'None', ensuring statistical alignment across all 700 transactions without removing data points.

3. Exploratory Data Analysis (EDA) Summary:

- * **Revenue Architecture:** Cross-sectional categorization reveals high-performing product segments and distinct geographical clusters that dominate total revenue volume.
- * **Temporal Patterns:** Historical transaction logs show explicit fluctuations based on month-over-month consumer demands, establishing baseline seasonal shifts.

4. Machine Learning & Predictive Modeling:

- * Model Selection: Implemented a Linear Regression Regressor utilizing categorical variables transformed via One-Hot encoding alongside transactional metrics like Units Sold, Sale Price, and Manufacturing Price.
- * Performance Evaluation Metrics:
 - Coefficient of Determination (R^2 Score): 0.7624 (The predictive framework effectively accounts for approximately 76.24% of the operational variance in global sales performance).
 - Root Mean Squared Error (RMSE): 114,707.06 (Establishes the average operational deviation threshold for forecasting outputs).

5. Strategic Recommendations for Stakeholders:

- * Optimize Promotional Strategy: Evaluate transactions tagged with high-tier discount bands to ensure that promotional strategies are actively scaling transaction frequency rather than diluting overall net profit margins.
- * Inventory Scaling: Align production logistics with top revenue-generating geographic regions and high-velocity product lines highlighted in the exploratory metrics to optimize inventory carrying costs.